

Effect of SiC particle size on the physical and mechanical properties of extruded Al matrix nanocomposites



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ABSTRACT

In this work, the effect of SiC particle size and its amount on both physical and mechanical properties of Al matrix composite were investigated. SiC of particle size 70 nm, 10 μm and 40 μm , and Al powder of particle size 60 μm were used. Composites of Al with 5 and 10 wt.% SiC were fabricated by powder metallurgy technique followed by hot extrusion. Phase composition and microstructure were characterized. Relative density, thermal conductivity, hardness and compression strength were studied. The results showed that the X-ray diffraction (XRD) analysis indicated that the dominant components were Al and SiC. Densification and thermal conductivity of the composites decreased with increasing the amount of SiC and increased with increasing SiC particle size. Scanning electron microscope (SEM) studies showed that the distribution of the reinforced particle was uniform. Increasing the amount of SiC leads to higher hardness and consequently improves the compressive strength of Al–SiC composite. Moreover, as the SiC particle size decreases, hardness and compressive strength increase. The use of fine SiC particles has a similar effect on both hardness and compressive strength.

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1. Introduction

Today, there is an increasing need worldwide for the advanced materials in order to obtain the desired properties. This is because a single material generally cannot meet the requirements of harsh engineering environments. That is why the need for composites with unique properties is growing every day. There is an increasing trend towards using composite materials in order to achieve better performance in engineering materials. Thus, production and application of metal matrix composites (MMCs) have increased in recent years [1].

During the last two decades, a lot of research has been conducted on aluminum matrix composites (AMCs). An optimum combination of high strength and ductility gives aluminum based metal matrix composites (AMCs) a wide range of possible advanced applications. The most important AMCs applications in aerospace and infrastructure industries are enabled by functional properties including high structural efficiency, excellent wear resistance, and attractive thermal and electrical characteristics [2–5].

Low wear resistance of pure aluminum is a serious drawback in using it in many applications. Addition of ceramic particles to aluminum matrix would improve the strength, hardness, and both the wear and corrosion resistance of the matrix [6,7]. Particle rein-

forcements are more favorable than fiber type, due to the better control of microstructure and mechanical properties, by varying the size and the volume fraction of the reinforcement [8]. SiC particles have been mostly selected as the reinforcement phase [9,10], this is due to, the high wear resistance, superior high temperature mechanical properties and low cost. Moreover, SiC does not react with the matrix at high temperatures and does not create undesired phases [11]. Challenging technical issues are overcome including compatibility between reinforcement and matrix, affordable primary and secondary processing techniques capable of adequately controlling reinforcement distribution, engineering design methodologies, and characterization and control of interfacial properties [12].

Powder metallurgy is considered as a good technique in producing metal–matrix composites. An important advantage of this method is its low processing temperature compared to melting techniques. Therefore, interaction between the matrix and the reinforcement phases is prevented. On the other hand, good distribution of the reinforcing particles can be achieved [13]. Another advantage of powder metallurgy technique is in its ability to manufacture near net shape product with low cost [14]. Extrusion can be applied from the beginning of the production or as a supplementary step after compaction.

It is commonly recognized that both the particle amount and particle size have obvious effects on the mechanical properties of the composites. The voids coexisted with the clustered SiC particles and around the large sized SiC particles can be treated

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as pre-existed cracks [15,16]. These particles cannot transfer any load from soft matrix to the hard reinforcements, and thus, lead to the degraded mechanical properties of the composites. Hardness, compression strength and wear resistance of the composite is improved by proper addition of the reinforcement. However, elongation of the composite may be reduced [8,17]. Reducing the particle size of the hard phase, greatly improves the strength which in turn depends on the manipulation method [18].

This work aims at studying the effect of particle size and amount of SiC on the physical and mechanical properties of Al. The prepared samples were investigated by XRD and the microstructure was studied by SEM. The relative density, thermal conductivity and compression strength of the samples were studied. The idea of performing compression test was taken from the application of such parts under compressive loads such as upsetting, the extent to which a material can be deformed in special metalworking process without the initiation of cracks.

2. Experimental work

2.1. Composite preparation

Aluminum powder with an average particle size of 60 μm and 99.9% purity was used as the matrix material (Fig. 1a). SiC powder with an average particle size of 70 nm was supplied from Alfa Aesar. Fig. 1b shows a transmission electron microscopy (TEM) picture of the particles. Micrometric SiC particles with two sizes of 10 μm and 40 μm were purchased from Alfa Aesar. Scanning electron microscopy (SEM) model a JEOL JSM- 5410, revealed the angular shape of the micrometric ceramic particles (Fig. 1c and d). The powders were mixed using a planetary ball mill for an hour, with the speed of 150 rpm. The ball to powder ratio was 10:1 and liquid ethanol was used as a process control agent. Specimens were prepared through powder metallurgy technique at a compaction pressure of 500 MPa, then sintered at 570 $^{\circ}\text{C}$ in a vacuum furnace with a heating rate of 10 $^{\circ}\text{C}$ per minute. The specimens were soaked at

420 $^{\circ}\text{C}$ for 1 h, 470 $^{\circ}\text{C}$ for another hour and finally at 570 $^{\circ}\text{C}$ for 5 h. After sintering, the specimens were hot extruded at 550 $^{\circ}\text{C}$ with an extrusion ratio of 9:1.

2.2. Composite characterization

The extruded samples were examined by X-ray diffraction (XRD) using Cu K α radiation with a wavelength of 1.5418 Å over a 2θ range of 20–90 $^{\circ}$. True densities of the composites were measured by using the Archimedes' method (ASTM: B962-13) and compared with theoretical densities to obtain varying degree of densification. The thermal conductivity can be calculated according the Wiedemann–Franz Law. Previous publication by the authors give a detailed description of the procedures [19].

Microstructure studies on the prepared composites were performed with a JEOL JSM-5410 model scanning electron microscope (SEM). In order to detect the Al, SiC and any oxide of Al and SiC particles, energy dispersive spectroscopy (EDS) analyses were performed. Hardness of both pure aluminum and composites were determined using Vickers indentation technique with a load of 3 kg at room temperature. Compressive testing was conducted using a universal testing machine, HU-F500KNI, according to standard ASTM: E9-09, at the room temperature and under a speed of 0.5 mm/min [20 or 21]. The compression test specimens were cylindrical with a height of 9 mm and gauge diameter of 9 mm. The specimen faces were polished with sandpaper up to grade 600 and then lubricated with graphite. The compression experiments were terminated at 60% specimen strains.

3. Results and discussion

3.1. X-ray diffraction

Fig. 2 shows X-ray diffraction (XRD) patterns of the extruded aluminum composite reinforced with different SiC particle size and amount. The XRD pattern shows two peaks that were identi-

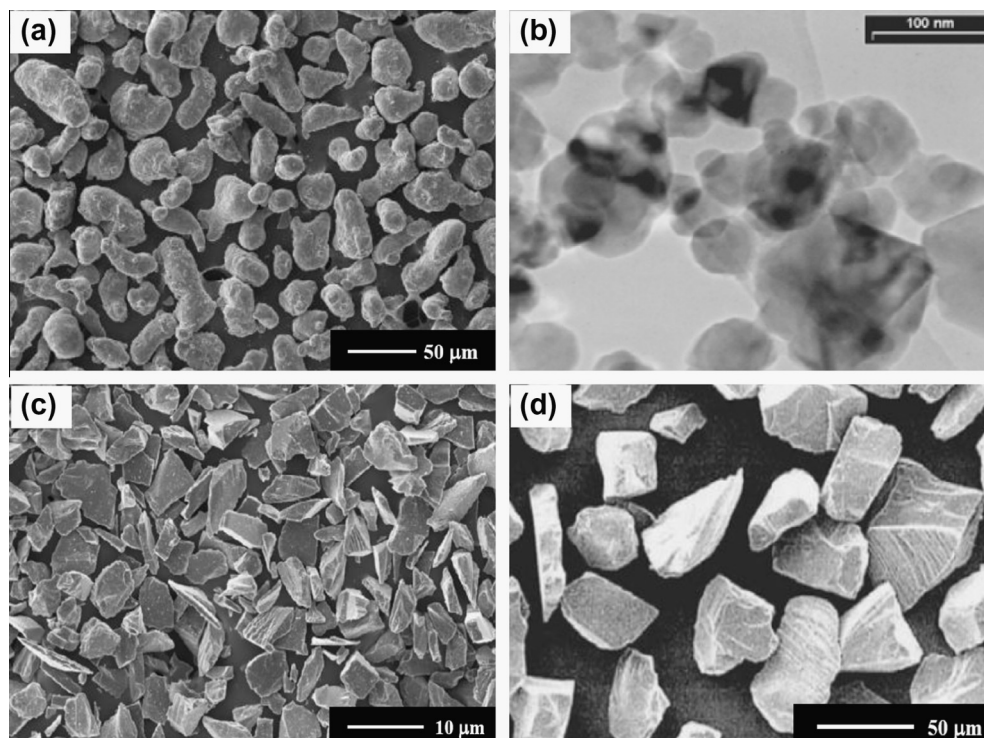


Fig. 1. Electron micrographs of starting powders of: (a) Al powder with 60 μm particle size, (b) SiC powder with 70 nm particle size, (c) SiC powder with 10 μm particle size and (d) SiC powder with 40 μm particle size.

fied as Al and SiC. XRD patterns of the composites with SiC particles of different sizes are similar to each other and consist of Al and SiC peaks dominantly. Furthermore, XRD peak intensities of the SiC phase are noticeably increased with increasing weight percentage and particle size of SiC. No oxide peak was observed in the XRD analyses of Al–SiC composites after extrusion. This suggests that no solid-state reactions between Al and SiC during sintering and extrusion processes. This result is very important as far as the electrical conductivity point of view is concerned for related materials.

3.2. The effect of SiC particle size and its amount on the densification

Fig. 3 shows the effect of SiC particle size and amount on the densification of extruded samples. This figure shows that as the amount of SiC increases, the densification declines. This is due to the agglomeration of SiC particles during the sintering process with increasing the SiC percent. It can be expected that in the sintering process the SiC particles are very dense with random distribution due to its high melting point. The dense network formed of SiC particles prevents to be dense [22]. In addition, the decline in the pressing capacity of samples with increasing in the amount of SiC is due to the higher hardness of SiC. Therefore, these composites have lower compressibility that results in lower densification.

Moreover, increasing the SiC particle size will increase the composite's densification. Probably, increasing the particle size of SiC leads to contact of more Al–Al grains together and so more intensive form appears [23]. By increasing the SiC particle size from 70 nm to 40 μm , composite densification increases. This is because specific surface of coarser particles is lower and the powder compressibility is higher. Hence, the matrix powder size has to be taken into account when dealing with density. When the alumi-

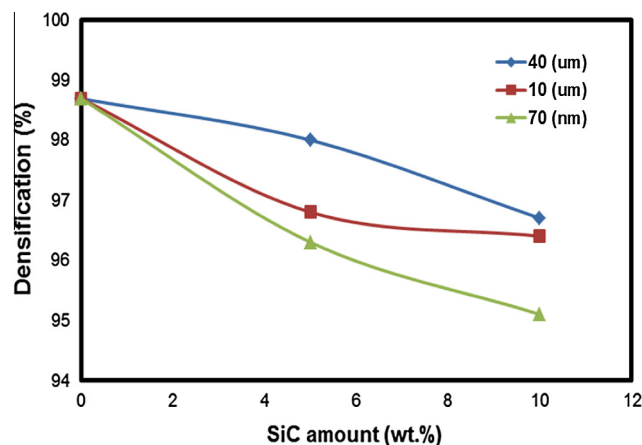


Fig. 3. Variation of the densification as a function of SiC particle size and amount.

num particle size is 60 μm , SiC particles of 70 nm, 10 μm and 40 μm are placed in the empty spaces among aluminum particles. Since the compressibility of pure aluminum is higher than that of its composite, the densification will be higher.

3.3. The effect of SiC particle size and its amount on the thermal conductivity

Fig. 4 shows the thermal conductivity of Al and Al–SiC composites depends on the particle size and amount of SiC. Thermal conductivity of composites decreases with increasing weight percentage of SiC. Thermal conductivity of a metal mainly depends on the motion of electrons in the structure. Ceramic-based SiC particles distort the structure and forms barrier for aluminum electrons providing conductivity. Therefore, conductivity of composites decreases with increasing the amount of SiC [24–26].

On the other hand, particle size of SiC also affects the thermal conductivity of composite and it increase as particle size increases. Thermal conductivity of composites increased with increasing particle size of SiC. Because with increasing particle size of SiC, added into the pure aluminum, fewer electrons are scattered and thus mobility and thermal conductivity increase [26]. This result is in very good agreement with study of Hussain et al. [27]. The conductivity of two-phase composites is determined by many factors, such as: (i) the conductivities of the constituent phases; (ii) the volume fractions and distributions of the constituent phases; (iii)

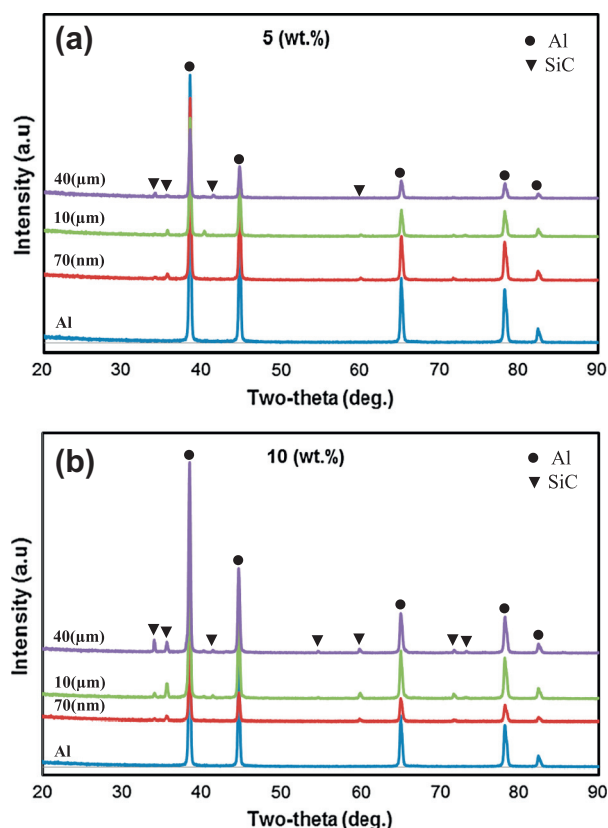


Fig. 2. XRD pattern of: (a) Al–5 wt.% SiC composite and (b) Al–10 wt.% SiC composite.

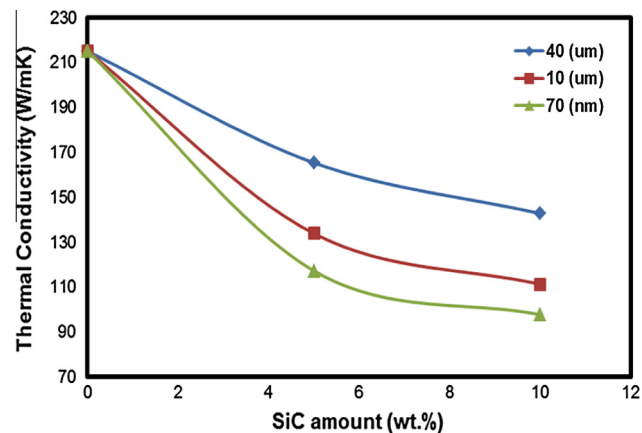


Fig. 4. The effect of SiC particle size and amount on thermal conductivity.

the size, shape, orientation and spacing of the phases; (iv) interaction between phases; and (v) the preparation method.

3.4. The effect of SiC particle size and its amount on the microstructure

The microstructure and mechanical properties of a composite is directly proportional to the amount and size of its reinforcements. Fig. 5 shows the photomicrographs of the samples containing 5 and 10 wt.% of SiC with different particle size of 70 nm, 10 μm and 40 μm . Grey regions imply Al matrix and dark grey and cornered particles imply the reinforcement component of SiC. It can be observed that SiC particles are well dispersed and uniformly distributed in Al matrix. The uniform distribution of SiC particles can be attributed to the time and the method of mixing. For the composite materials, it is very important to obtain homogeneous reinforcement in the matrix to enhance the mechanical, electrical and thermal properties [28].

Also it can be observed that the increasing of the amount of SiC leads to a finer grain sizes. This is because SiC particles prevent the grain growth during the sintering process. In addition, figure depicts that as reinforcement particle size increases, the grain size will also increase. According to Eq. (1), as the reinforcement parti-

cle size increases, the distance between the particles will also increase [29].

$$\lambda = \frac{4(1-f)r}{3f} \quad (1)$$

where λ is the distance between the reinforcement particles, f is the particles volume fraction, and r is the particle radius, assuming them as spherical. Since SiC particles act as obstacles against grain boundary movement at the time of sintering, increasing the distance between particles (due to coarse-grained SiC), decreases the number of obstacles against grain boundary movement. As a result, the grains move freely and the grain size increases.

3.5. The effect of SiC particle size and its amount on the hardness

Fig. 6 shows the effect of the SiC particle size and amount on the composite hardness. Considering this figure, by increasing the amount of SiC, composite hardness increases since its hardness is much higher than that of pure aluminum [30]. This fact can be easily analyzed through the rule of mixtures.

$$H_c = H_m f_m + H_r f_r \quad (2)$$

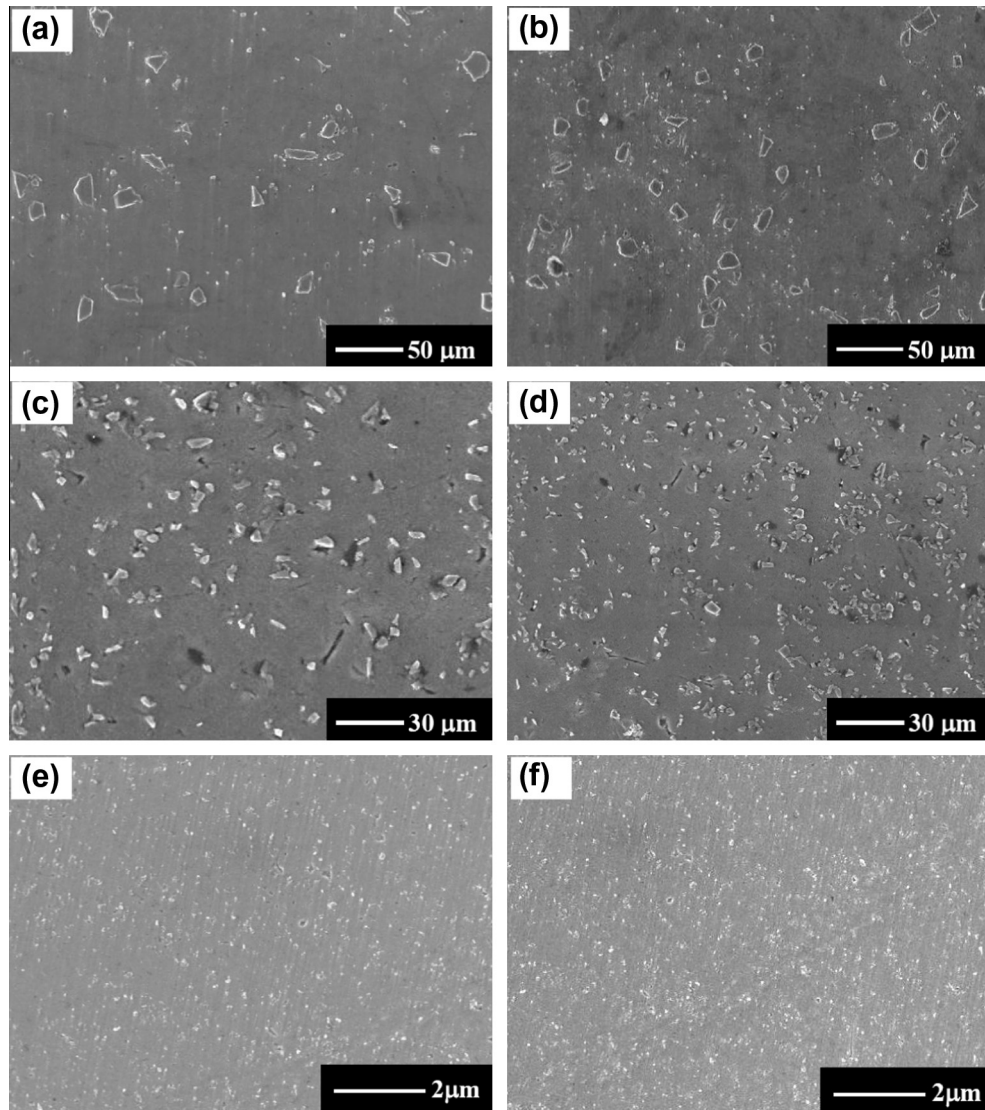


Fig. 5. SEM micrographs of extruded composite: (a) Al–5 wt.% SiC (40 μm), (b) Al–10 wt.% SiC (40 μm), (c) Al–5 wt.% SiC (10 μm), (d) Al–10 wt.% SiC (10 μm), (e) Al–5 wt.% SiC (70 nm) and (f) Al–5 wt.% SiC (70 nm).

H_r , H_m and H_c are the hardness of reinforcement, matrix and composite, where f_r and f_m are volume fraction of reinforcement and matrix, respectively. It was observed from this study that by increasing the amount of SiC from 0% to 10%, hardness increased from 40.3 to 52.8 HV. Apparently, the hardness of aluminum improves considerably with the additions of SiC particles at the expense of its ductility that can be attributed to higher hardness of SiC. This result was consistent with another research [31]. It is thought that higher amount of ceramic particles in the matrix would result in more dislocations that increases the hardness of the composite [32–34].

Considering this figure also, it shows that decreasing the particle size will increase the composite hardness. Therefore, the average hardness of Al–SiC composite specimens with 70 nm SiC particles is higher than that of the Al–SiC composite specimens with 10 and 40 μm SiC particles. The reason for this increase can be examined from two perspectives. One is due to greater interfacial area between the hard and soft phases [35]. Secondly, the defects in the coarse-grained particles are more than the fine-grained ones which results in its easy fracture under tension.

3.6. The effect of SiC particle size and its amount on the compressive strength

Fig. 7 shows the stress–strain curves from compression tests of aluminum composite reinforced with different SiC particle size and amount after extrusion. A considerable difference is seen between the strength of pure aluminum sample and the composite samples. It is shown that the sample containing 10 wt.% SiC has generally better strength than the 5 wt.% SiC and pure aluminum samples.

On the other hand, the observed changes in the strength may be attributed to changes in the particle size of SiC. A close look at the diagrams shows that the samples containing 70 nm SiC particles have comparatively exceptional properties. Additionally, the stress–strain behavior of the sample containing 40 μm particle sizes is different from two other samples. The reason is due to the great difference in their particle size. Considering these diagrams, the sample containing 10 wt.% SiC, 70 nm particle size, has the highest amount of compressive strength which is 601 MPa.

Although tensile testing can determine if a proper bonding has been produced, compression tests can also be helpful to study the barreling and upsetting behavior of the compacts. Stronger bonds among particles delay the crack initiation during deformation especially at the time of barreling [36,37].

Fig. 8 illustrates the effect of the SiC particle size and amount on the compressive strength at a strain of 0.6. The addition of up to 10 wt.% SiC to pure aluminum powder increases the compressive strength. The reason is that SiC particles prevent the movement

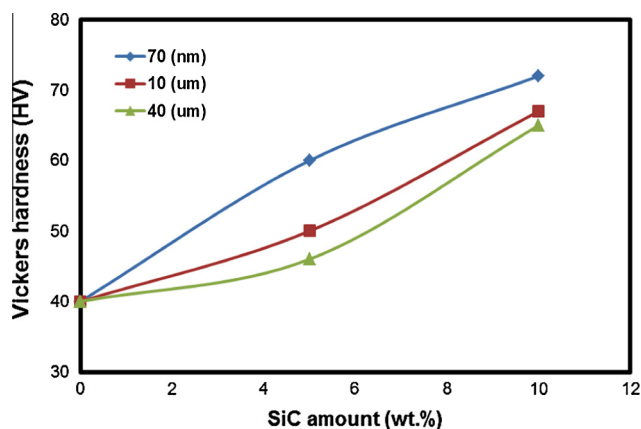


Fig. 6. The effect of SiC particle size and amount on composite hardness.

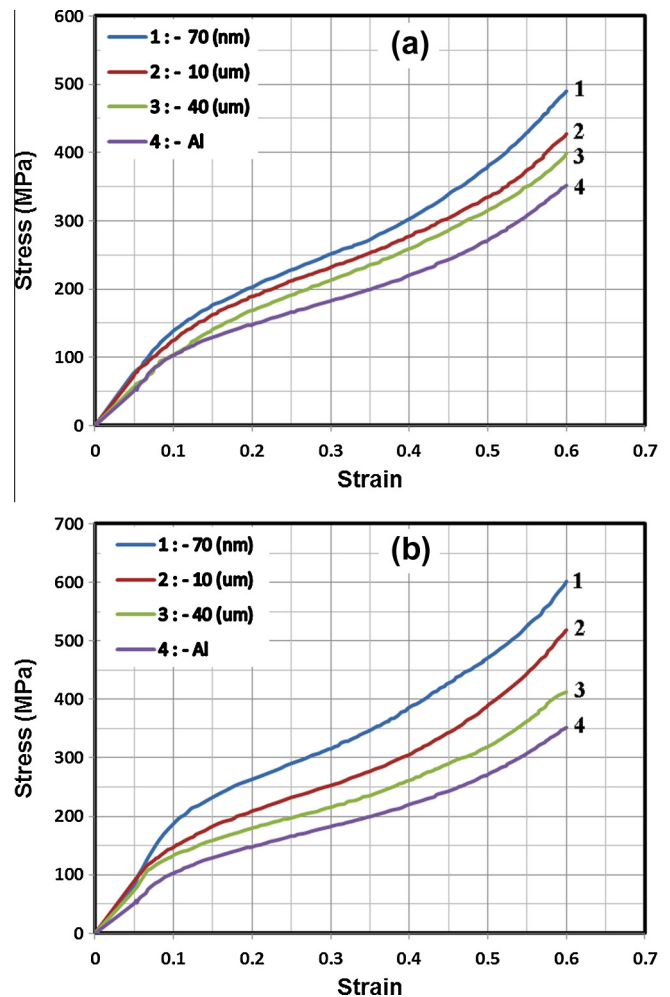


Fig. 7. The effect of SiC particle size on the stress–strain behavior of the composites: (a) Al–5 wt.% SiC and (b) Al–10 wt.% SiC.

of dislocations in pure aluminum matrix through dispersion strengthening mechanism [29]. Additionally, according to Eq. (1), increasing the amount of SiC leads to a decrease in the distance between the SiC particles. Decreasing the distance between the SiC particles according to Eq. (3) will increase the required tension for dislocations movement between the SiC particles leading to an increase in the material strength.

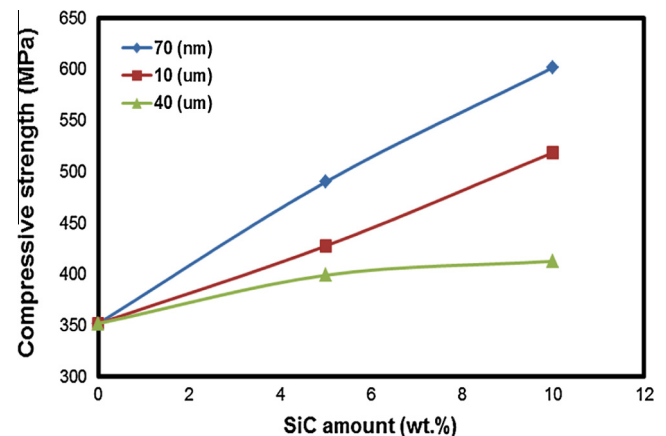


Fig. 8. The variation of the compressive strength of the composites as a function of SiC particle size and content.

$$\tau_0 = \frac{Gb}{\lambda} \quad (3)$$

In Eq. (3), τ_0 is the required tension for forcing dislocations to move among reinforcement particles. G is the material's elastic modulus and b is the Berger's vector [29].

Considering Fig. 8, decreasing the SiC particle size will increase the strength. The reason for such property may be attributed to Eq. (1). Lowering the reinforcement particle size at a constant amount decreases the distance between the reinforcement particles. Additionally, according to Eq. (3), decreasing the distance between the reinforcement particles increases the required tension for moving the dislocations among the reinforcements, leading to increased composite strength [29].

This is in agreement with hardness values mentioned earlier. This may be explained by the Hall–Petch relationship.

$$\sigma_o = \sigma_i + KD^{-1/2} \quad (4)$$

In this equation, σ_o is the flow stress, σ_i is the stress opposing the movement of dislocations, K is constant and D , is the grain size. According to Eq. (4), as the grain size gets smaller, flow stress also increases, leading to high strength in the composite [29].

4. Conclusions

The effects of particle size and amount of reinforcement component on the some properties of extruded Al–SiC nanocomposites were investigated. The following findings can be summarized:

1. The presence of Al and SiC were verified by XRD analysis technique.
2. As the amount of SiC particles increase, the composite's densification declines. Also, as the SiC particle size increases in the composite samples, densification increases.
3. Increasing the amount and reducing the size of SiC lead to reduced thermal conductivity.
4. According to SEM micrographs, as the SiC particles size decreases, the grain size and the distribution of reinforcement decreases. Furthermore, as the SiC amount decreases in the composite, grain size and SiC particle distribution homogeneity increases.
5. Increasing the amount and reducing the size of SiC promote high hardness in the composite. The highest hardness of 72 HV is attributed to the sample containing 10 wt.% of SiC with 70 nm particle size.
6. The finer particle size of SiC presents greater compressive strength. By increasing the amount of SiC, compressive strength increases. The highest strength was 601 MPa, for the composite containing 10 wt.% SiC with 70 nm.

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